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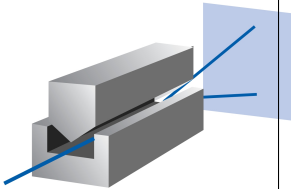
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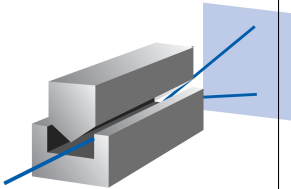
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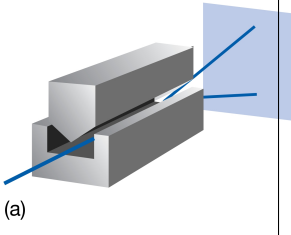
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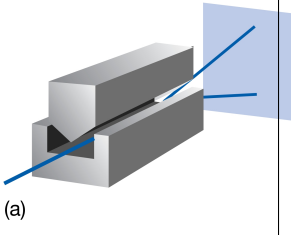


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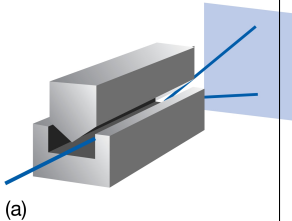
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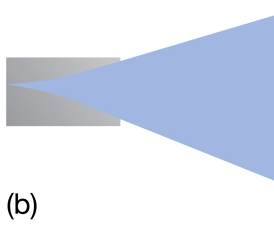
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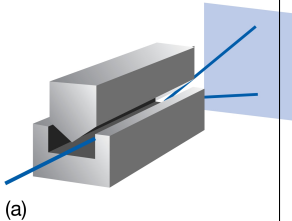
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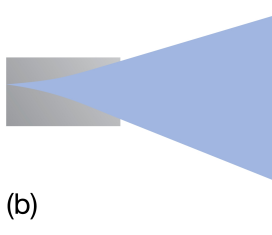
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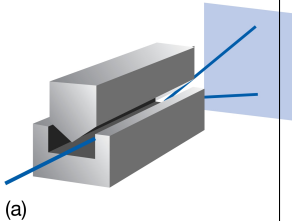


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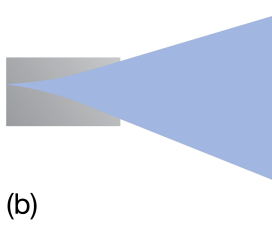
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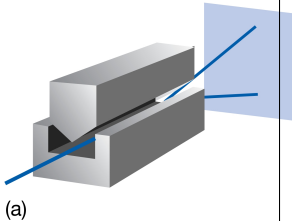


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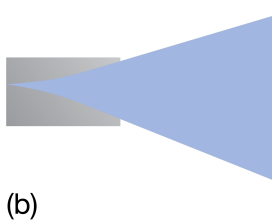
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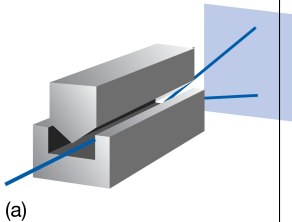


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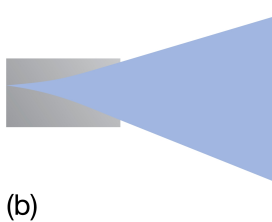
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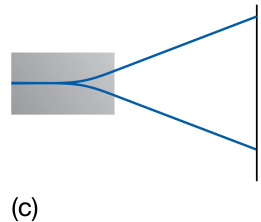
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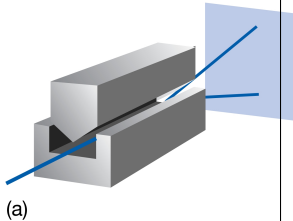
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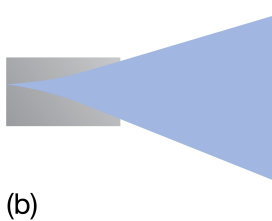
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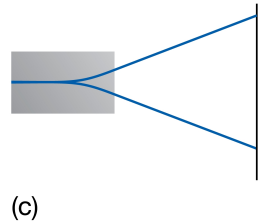
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Expt. repeated with H atoms in 1927

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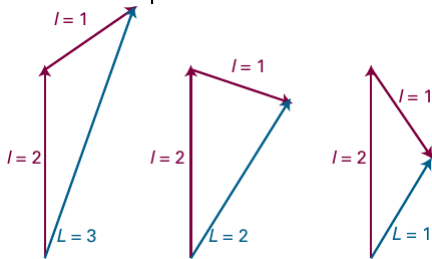
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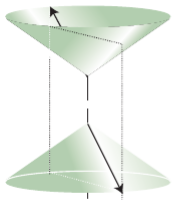
Ex.(i) for 2 spin-less particles, $\hat{\vec{L}} = \hat{\vec{L}}_1 + \hat{\vec{L}}_2$ where $\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$ = orbital angular momentum operator

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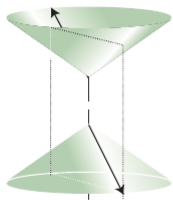
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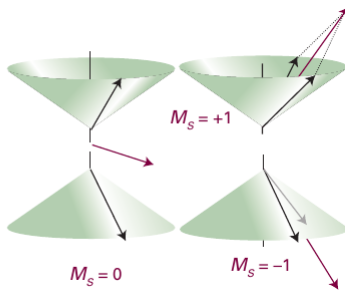


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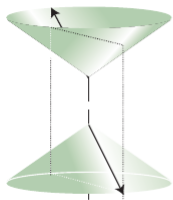


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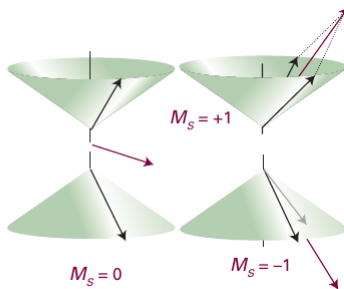


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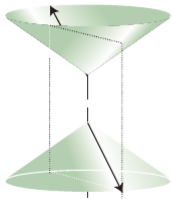


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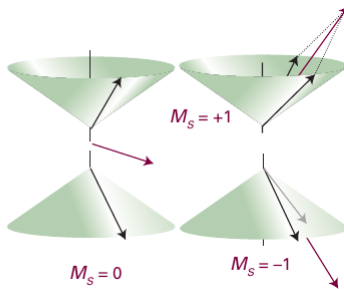


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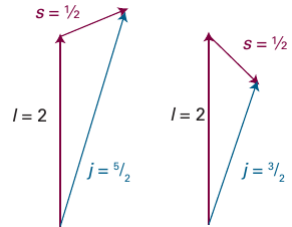
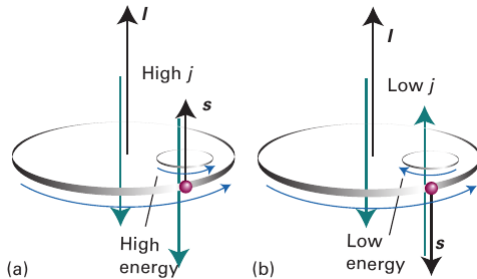


(a) $S = 0$
singlet



(b) $S = 1$
triplet

Ex. (iii) for single particle, $\hat{J} = \hat{L} + \hat{S}$, where $\hat{L} = \hat{r} \times \hat{p}$ = orbital angular momentum operator and $\hat{S}$ = spin angular momentum operator



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$$\hat{J}_i^2 |j, M, j_1, j_2\rangle = j_i(j_i + 1)\hbar^2 |j, M, j_1, j_2\rangle$$

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Coeffs depend on j, M, j_1, j_2, m_1 and m_2 : Clebsch Gordan (C. G.) coeffs

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Another ex.: $j_2=2$
 $j_1=1$

$$|j_1+j_2 \underbrace{M_{\max}}_{j_1+j_2} j_1 j_2\rangle = |j_1 \underbrace{m_{1\max}}_{j_1} j_2, \underbrace{m_{2\max}}_{j_2}\rangle \quad |j=2+1, 3, 1, 2\rangle = |1, 1, 2, 2\rangle$$

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$$|j, -3, 1, 2\rangle = |1, -1, 2, -2\rangle$$

Max and min values of M correspond to $j = j_1 + j_2$,

$$\text{s. t. } M_S \in \left\{ \underbrace{-j}_{-(j_1+j_2)}, \dots, \underbrace{j}_{j_1+j_2} \right\} \text{ in increments of 1}$$

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However, z-components add to $\frac{1}{2}\hbar + \frac{1}{2}\hbar = \hbar$

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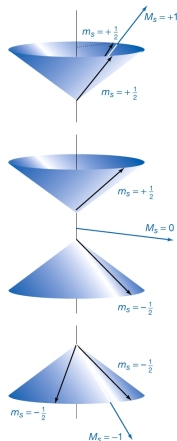
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$$\implies ac + bd = 0, \text{ or, } \frac{1}{\sqrt{2}}(c + d) = 0, \text{ or, } c = -d \quad ; \text{ normalisation: } c = -d = \frac{1}{\sqrt{2}}$$

For $S = 1$, there are $2S + 1 = 3$ states. This is triplet combination.

All these are **symmetric** w.r.t. exchange of particles 1 and 2:

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Exercise: Use $\widehat{S}_-$ on $|S = 1, M_s = 0\rangle$ to show that $|S = 1, M_s = -1\rangle \equiv |11\rangle$

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$|2\ 2\rangle_{II} \equiv |1\ 1\rangle_I$ and $|2\ -2\rangle_{II} \equiv |-1\ -1\rangle_I$

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$$|1 \ 1\rangle_{II} = a|0 \ 1\rangle_I + b|1 \ 0\rangle_I$$

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This must be orthogonal to $|2 1\rangle_{II}$. This gives, $a = -b$. Then, using normalisation, we have, $a^2 + b^2 = 1$ and $a = -b = \frac{1}{\sqrt{2}}$

$$\therefore |1 1\rangle_{II} = \frac{1}{\sqrt{2}} (|0 1\rangle_I - |1 0\rangle_I)$$

Exercise: Verify that

$$|2 - 1\rangle_{II} = \frac{1}{\sqrt{2}} [|0 - 1\rangle_I + |- 1 0\rangle_I]$$

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$$\text{Now, } \widehat{J_-}|1 1\rangle_{II} = \frac{1}{\sqrt{2}} \left[\widehat{J_{1-}}|0 1\rangle_I + \widehat{J_{2-}}|0 1\rangle_I - \widehat{J_{1-}}|1 0\rangle_I - \widehat{J_{2-}}|1 0\rangle_I \right]$$

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$$\text{Proceed to prove: } |1 0\rangle_{II} = \frac{1}{\sqrt{2}} (|-1 1\rangle_I - |1 -1\rangle_I)$$

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$$\text{Now, } \widehat{J_-}|1 1\rangle_{II} = \frac{1}{\sqrt{2}} \left[\widehat{J_{1-}}|0 1\rangle_I + \widehat{J_{2-}}|0 1\rangle_I - \widehat{J_{1-}}|1 0\rangle_I - \widehat{J_{2-}}|1 0\rangle_I \right]$$

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$$\text{and then } |1 -1\rangle_{II} = \frac{1}{\sqrt{2}} (|0 -1\rangle_I - |-1 0\rangle_I)$$

We have all the C.G.s for $j = 1$.

Finally, consider $j = 0$,

$$|0\ 0\rangle_{II} = a|1\ -1\rangle_I + b|0\ 0\rangle_I + c|-1\ 1\rangle_I$$

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This gives two equations:

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This must be orthogonal to $|2\ 0\rangle_{II}$ and $|1\ 0\rangle_{II}$.

This gives two equations:

$$a + 2b + c = 0; \text{ and } a = c \quad \implies b = -c$$

Normalisation gives,

$$a^2 + b^2 + c^2 = 1 \implies 3c^2 = 1 \implies c = a = -b = \frac{1}{\sqrt{3}}$$

$$\therefore |0\ 0\rangle_{II} = \frac{1}{\sqrt{3}} [|1\ -1\rangle_I - |0\ 0\rangle_I + |-1\ 1\rangle_I]$$

$$j_1 = 1 ; j_2 = 1 :$$

$$\text{We can write as } \underbrace{\langle m_1 m_2 }_I \underbrace{| j M \rangle}_{II} =$$

$$\langle 1 \quad 1 | 2 \quad 2 \rangle = 1$$

$$\langle 1 \ 0 | 2 \ 1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 0 \ 1 | 2 \ 1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 0 \ 0 | 2 \ 0 \rangle = \frac{2}{\sqrt{6}}$$

$$\langle 1 \quad -1 | 2 \ 0 \rangle = \frac{1}{\sqrt{6}}$$

$$\langle -1 \ 1 | 2 \ 0 \rangle = \frac{1}{\sqrt{6}}$$

$$\langle 0 \quad -1 | 2 \quad -1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle -1 \ 0 | 2 \quad -1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle -1 \quad -1 | 2 \quad -2 \rangle = 1$$

$$\text{or with more detail, } \langle j_1 m_1 j_2 m_2 | j M \rangle =$$

$$\langle 1 \ 1 \ 1 \ 1 | 2 \ 2 \rangle = 1$$

$$\langle 1 \ 1 \ 1 \ 0 | 2 \ 1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 1 \ 0 \ 1 \ 1 | 2 \ 1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 1 \ 0 \ 1 \ 0 | 2 \ 0 \rangle = \frac{2}{\sqrt{6}}$$

$$\langle 1 \ 1 \ 1 \quad -1 | 2 \ 0 \rangle = \frac{1}{\sqrt{6}}$$

$$\langle 1 \quad -1 \ 1 \ 1 | 2 \ 0 \rangle = \frac{1}{\sqrt{6}}$$

$$\langle 1 \ 0 \ 1 \quad -1 | 2 \quad -1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 1 \quad -1 \ 1 \ 0 | 2 \quad -1 \rangle = \frac{1}{\sqrt{2}}$$

$$\langle 1 \quad -1 \ 1 \quad -1 | 2 \quad -2 \rangle = 1$$

$$\langle 1 \quad -1 \ 1 \ 1 | 1 \ 0 \rangle = \frac{1}{\sqrt{2}}$$

Breit Scheme: Obtaining states for identical electrons:

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Ex. p^2 :

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	Each electron	electrons 1 and 2
Ex. p^2 :	$l = 1; 2l + 1 = 3$ $s = \frac{1}{2}; 2s + 1 = 2$	$3 \times 2 = 6; 6^2 = 36$ states

Breit Scheme: Obtaining states for identical electrons:

	Each electron	electrons 1 and 2
Ex. p^2 :	$l = 1; 2l + 1 = 3$ $s = \frac{1}{2}; 2s + 1 = 2$	$3 \times 2 = 6; 6^2 = 36$ states Can also be seen as: $(2l + 1)^2(2s + 1)^2 = 3^2 \cdot 2^2 = 36$

After coupling,

$$\begin{array}{l|l}
 L = \{1 + 1, 1 + 1 - 1, 1 - 1\} \equiv \{2, 1, 0\} & {}^3D, {}^3P, {}^3S \\
 & 3 \times 5 = 15 \\
 & 3 \times 3 = 9 \\
 & 3 \times 1 = 3 \\
 S = \{\frac{1}{2} + \frac{1}{2}, \frac{1}{2} - \frac{1}{2}\} \equiv \{1, 0\} & {}^1D, {}^1P, {}^1S \\
 & 5 \\
 & 3 \\
 & 1 \\
 & \text{total}=36
 \end{array}$$

In case of equivalent electrons, Pauli's exclusion principle is invoked.

	$m_l =$	1	0	-1	1	0	-1
Consider for each electron	$m_s =$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$
		a	b	c	d	e	f

Total possibilities $\Rightarrow \binom{6}{2} = 15$.

[illegible]

	ab	ac	ad	ae	af	bc	bd	be	bf	cd	ce	cf	de	df	ef
M_L	1	0	2	1	0	-1	1	0	-1	0	-1	-2	1	0	-1
M_S	1	1	0	0	0	1	0	0	0	0	0	0	-1	-1	-1

Construct matrices:

m_{l_1}	1	0	-1	
M_L	2	1	0	1
	1	0	-1	0
	0	-1	-2	-1
	S	P	D	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

like spin z-components (m_s) $\implies M_S = 1$ or -1

diagonal elements of M_L matrix not allowed

symmetric states from upper right taken (identical)

$$M_S = 1 \quad M_L = 1 \quad 0 \quad -1 \quad (L = 1, {}^3P)$$

$$-1 \quad 1 \quad 0 \quad -1 \quad (L = 1, {}^3P)$$

two possibilities ac and df for $M_s = \pm 1$ and $M_L = 0$ are already accounted

so lower left is ignored

	ab	ac	ad	ae	af	bc	bd	be	bf	cd	ce	cf	de	df	ef
M_L	1	0	2	1	0	-1	1	0	-1	0	-1	-2	1	0	-1
M_S	1	1	0	0	0	1	0	0	0	0	0	0	-1	-1	-1

Construct matrices:

m_{l_1}	1	0	-1		m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_L	2	1	0	1	M_S	1	0	$\frac{1}{2}$
	1	0	-1	0		0	-1	$-\frac{1}{2}$
	$\emptyset$	-1	-2	-1		Singlet	Triplet	
	S	P	D	m_{l_2}				m_{s_2}

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$M_S = \pm 1$ comes from a triplet, together with $M_S = 0$.

	ab	ac	ad	ae	af	bc	bd	be	bf	cd	ce	cf	de	df	ef
M_L	1	0	2	1	0	-1	1	0	-1	0	-1	-2	1	0	-1
M_S	1	1	0	0	0	1	0	0	0	0	0	0	-1	-1	-1

Construct matrices:

m_{l_1}	1	0	-1	
M_L	2	1	0	1
	1	0	-1	0
	0	-1	-2	-1
	S	P	D	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

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two possibilities ac and df for $M_s = \pm 1$ and $M_L = 0$ are already accounted

so lower left is ignored

$M_S = \pm 1$ comes from a triplet, together with $M_S = 0$. Where is $M_S = 0$?

m_{l_1}	1	0	-1	
M_L	2	1	0	1
	1	0	-1	0
	0	-1	-2	-1
	S	P	D	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

If m_l like then diagonal in M_S matrix not allowed

m_{l_1}	1	0	-1	
M_L	2	1	0	1
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	0	-1	-2	-1
	S	P	D	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

If m_l like then diagonal in M_S matrix not allowed

$$M_S = 0; \quad M_L = \begin{matrix} & 2 & 1 & 0 & -1 & -2 \\ & 1 & 0 & -1 & & \\ & & 0 & & & \end{matrix} \quad \begin{matrix} (L=2, {}^1D) \\ M_S = 0 \text{ from triplet} \\ (L=0, {}^1S) \end{matrix}$$

Total number of states: $\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 2 & 1 \end{matrix}$

Having accounted for the triplet $M_S = 0$, we see that no 1P is possible

Going back to the table above, we find,

$$M_L = \begin{matrix} 2 & 1 & 0 & -1 & -2 \\ 1 & 0 & -1 & & \\ 0 & -1 & -2 & & \\ & & & & \end{matrix} \Rightarrow {}^1D \text{ (5 states)}$$

$$M_S = \begin{matrix} 0 & 0 & 0 & 0 & 0 \\ & 1 & 0 & -1 & \\ & 1 & 1 & 1 & \\ & 0 & 0 & 0 & \\ & -1 & -1 & -1 & \end{matrix} \Rightarrow {}^3P \text{ (9 states)}$$

$$\text{and } M_L = \begin{matrix} 0 \\ 0 \end{matrix} \Rightarrow {}^1S \text{ (1 state); total = 15}$$

$$\begin{array}{rcccccc} M_L = & 2 & 1 & 0 & -1 & -2 \\ M_S = & 0 & 0 & 0 & 0 & 0 \end{array} \Rightarrow {}^1D \text{ (5 states)}$$

$$\begin{array}{rcccc} M_L = & 1 & 0 & -1 \\ M_S = & 1 & 1 & 1 \\ & 0 & 0 & 0 \\ & -1 & -1 & -1 \end{array} \Rightarrow {}^3P \text{ (9 states)}$$

$$\text{and } \begin{array}{rcc} M_L = & 0 \\ M_S = & 0 \end{array} \Rightarrow {}^1S \text{ (1 state); total = 15}$$

d^2 : Total number of states = $(2l + 1)^2(2s + 1)^2 = 5^2 \cdot 2^2 = 100$

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$$\text{Non-equivalent: } L = \begin{Bmatrix} 2+2 \\ \vdots \\ \vdots \\ 2-2 \end{Bmatrix} = \begin{matrix} 4 \\ 3 \\ 2 \\ 1 \\ 0 \end{matrix} \quad S = \left\{ \begin{matrix} \frac{1}{2} + \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} \end{matrix} \right\} = \begin{matrix} 1 \\ 0 \end{matrix}$$

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states:	${}^3G(27)$ ${}^3F(21)$ ${}^3D(15)$ ${}^3P(9)$ ${}^3S(3)$	${}^1G(9)$ ${}^1F(7)$ ${}^1D(5)$ ${}^1P(3)$ ${}^1S(1)$	75 + 25 = 100
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Equivalent electrons:

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
S	P	D	F	G	m_{l_2}	

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
Singlet	Triplet		m_{s_2}

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	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

like spin z-components(m_s) $\implies M_S = 1$ or -1 :

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	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

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i) diagonal elements of M_L matrix not allowed

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m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

like spin z-components(m_s) $\implies M_S = 1$ or -1 :

i) diagonal elements of M_L matrix not allowed

ii) the lower left is ignored; identical states from upper half considered

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

$M_S =$	1		$M_L =$	3	2	1	0	-1	-2	-3
	-1			3	2	1	0	-1	-2	-3
	1					1	0	-1		
	-1					1	0	-1		

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

$M_S =$	1		$M_L =$	3	2	1	0	-1	-2	-3
	-1			3	2	1	0	-1	-2	-3
	1					1	0	-1		
	-1					1	0	-1		

once again $M_S = 0$ from the triplet is missing, we will find it later

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

$M_S =$	1		$M_L =$	3	2	1	0	-1	-2	-3
	-1			3	2	1	0	-1	-2	-3
	1					1	0	-1		
	-1					1	0	-1		

once again $M_S = 0$ from the triplet is missing, we will find it later
 alike m_l values: diagonal elements of $\{M_S\}$ matrix not allowed

m_{l_1}	2	1	0	-1	-2	
M_L	4	3	2	1	0	2
	3	2	1	0	-1	1
	2	1	0	-1	-2	0
	1	0	-1	-2	-3	-1
	0	-1	-2	-3	-4	-2
	S	P	D	F	G	m_{l_2}

m_{s_1}	$\frac{1}{2}$	$-\frac{1}{2}$	
M_S	1	0	$\frac{1}{2}$
	0	-1	$-\frac{1}{2}$
	Singlet	Triplet	m_{s_2}

$$\begin{array}{cccccccc}
 M_S = & 1 & & M_L = & 3 & 2 & 1 & 0 & -1 & -2 & -3 \\
 & -1 & & & 3 & 2 & 1 & 0 & -1 & -2 & -3 \\
 & 1 & & & & & 1 & 0 & -1 & & \\
 & -1 & & & & & 1 & 0 & -1 & &
 \end{array}$$

once again $M_S = 0$ from the triplet is missing, we will find it later

alike m_l values: diagonal elements of $\{M_S\}$ matrix not allowed

$$\begin{array}{cccccccccccc}
 M_S = 0 & & & & & & & & & & & & & \\
 M_L = & 4 & 3 & 2 & 1 & 0 & -1 & -2 & -3 & -4 \\
 & & 3 & 2 & 1 & 0 & -1 & -2 & -3 \\
 & & & 2 & 1 & 0 & -1 & -2 & -3 \\
 & & & & 1 & 0 & -1 \\
 & & & & & 0
 \end{array}$$

Total number of states	1	2	3	4	5	4	3	2	1
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